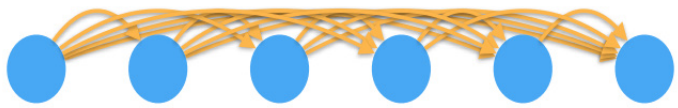
Lecture 3

Sequence Models 序列模型

① 时序模型：当前数据与之前观察到的数据相关

$$p(x) = p(x_1) \cdot p(x_2|x_1) \cdot p(x_3|x_1, x_2) \cdot \ldots \cdot p(x_T|x_1, \ldots x_{T-1})$$

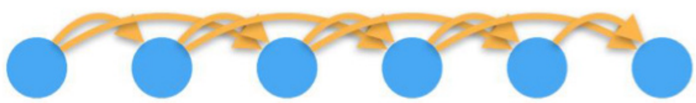


② 自回归模型：用自身过去数据预测未来

$$p(x_t|x_1, \ldots x_{t-1}) = p(x_t|f(x_1, \ldots, x_{t-1}))$$

③ Markov Models：假设当前数据只跟τ个过去数据点相关

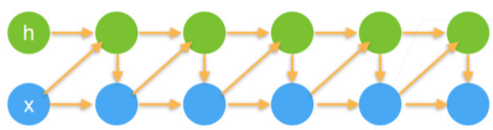
$$p(x_t|x_1, \ldots, x_{t-1}) = p(x_t|x_{t-\tau}, \ldots, x_{t-1}) = p(x_t|f(x_{t-\tau}, \ldots, x_{t-1}))$$



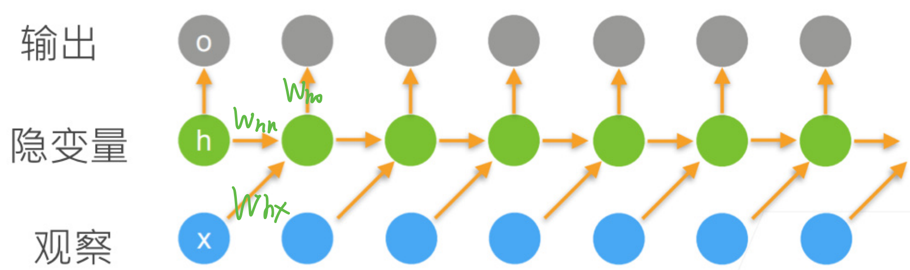
④ Hidden Variable Models：引入潜变量来表示过去信息 $h(t) = f(x_1, \ldots, x_{t-1})$

$p(h_t | h_{t-1}, x_{t-1})$

$p(x_t | h_t, x_{t-1})$



Recurrent Neural Networks → Language Models



更新隐藏状态：$\mathbf{h}_t = \phi(\mathbf{W}_{hh}\mathbf{h}_{t-1} + \mathbf{W}_{hx}\mathbf{x}_{t-1} + \mathbf{b}_h)$

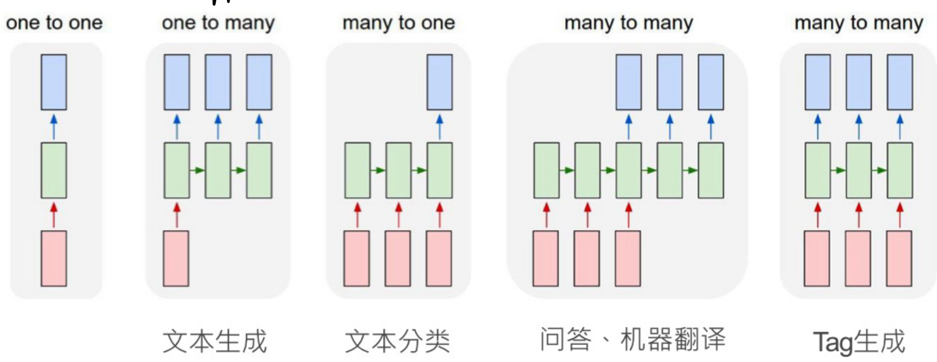
输出：$\mathbf{o}_t = \phi(\mathbf{W}_{ho}\mathbf{h}_t + \mathbf{b}_o)$

Evaluate a language Model：① Average Cross Entropy $\frac{1}{n}\sum_{t=1}^{n} -\log p(x_t|x_{t-1} \cdots)$

② perplexity $\exp(-\frac{1}{n}\sum_{t=1}^{n} \log p(x_t|x_{t-1}, \cdots, x_1))$

best：ppl=1 worst：ppl=∞

RNN's Application:



Gated Recurrent Units (GRU)

Activation function (sigmoid) 输出0/1 决定是否Reset

$R_t = \sigma(X_t W_{xr} + H_{t-1} W_{hr} + b_r)$ Reset Gate

Activation function

$Z_t = \sigma(X_t W_{xz} + H_{t-1} W_{hz} + b_z)$ Upload Gate.

逐元素相乘(哈达玛积)

$\tilde{H}_t = \tanh(X_t W_{xh} + (R_t \odot H_{t-1}) W_{hh} + b_h)$ Candidate Hidden State.

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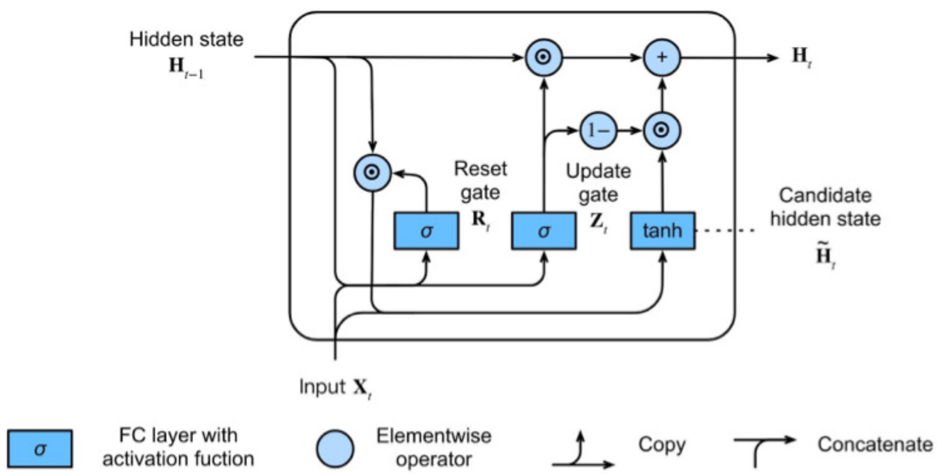
$R_t = 0$ 忘 $\tilde{H}_t$ 全来源于 X_t 不提取 H_{t-1} $R_t = 1$ 不忘 $\tilde{H}_t$ 由 H_{t-1} 与 X_t 共同决定

完全更新. $Z_t = 0$ H_t 完全来自新输入 完全不更新. $Z_t = 1$ H_t 完全来自上一时间步的 H_{t-1}

$$H_t = Z_t \odot H_{t-1} + (1 - Z_t) \odot \tilde{H}_t$$ Hidden State

更新门 更新门



Long Short Term Memory (LSTM)

sigmoid 0/1 决定是否

Gates: $I_t = \sigma(X_t W_{xi} + H_{t-1} W_{hi} + b_i)$ Input Gate

Activation function

$F_t = \sigma(X_t W_{xf} + H_{t-1} W_{hf} + b_f)$ Forget Gate

$O_t = \sigma(X_t W_{xo} + H_{t-1} W_{ho} + b_o)$ Output Gate

Cell : $\tilde{C}_t = \tanh(X_t W_{xc} + H_{t-1} W_{hc} + b_c)$ Candidate Memory Cell

克服梯度消失

控制忘多少 C_{t-1} 控制保存多少 $\tilde{C}_t$

忘 输入 $F_t=0$ $I_t=1$ 完全依赖候选状态 $\tilde{C}_t$

不忘 不输入 $F_t=1$ $I_t=0$ 完全依赖历史信息 C_{t-1}

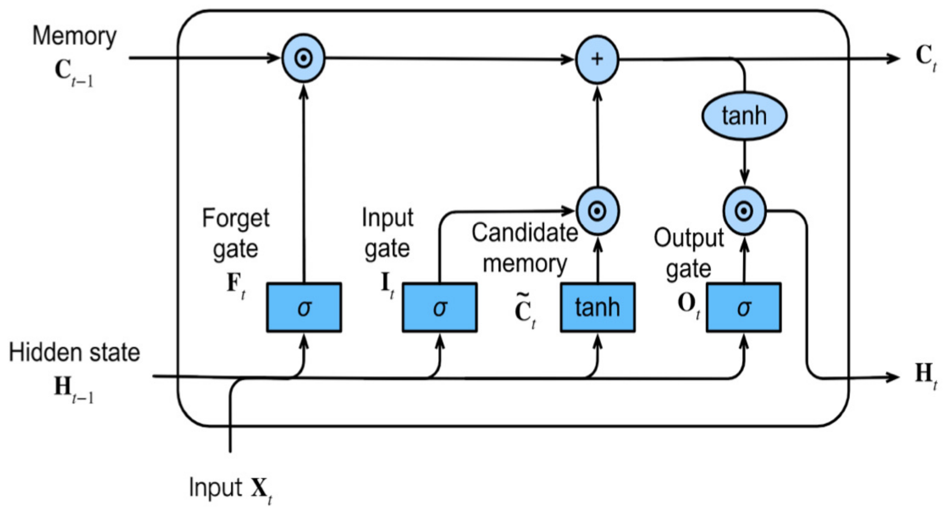
遗忘门 输入门

$C_t = F_t \odot C_{t-1} + I_t \odot \tilde{C}_t$ Memory Cell

$H_t = O_t \odot \tanh(C_t)$ Hidden State

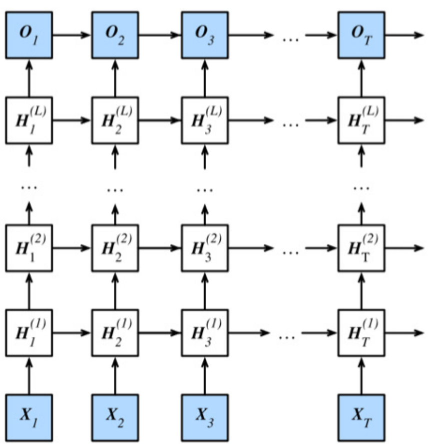
输出门

控制内部状态C_t要输出多少给外部状态h_t



Deep Recurrent Neural Networks 深度↗

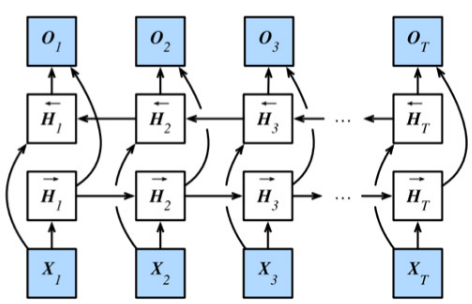
- 浅RNN
  - 输入
  - 隐层
  - 输出
- 深RNN
  - 输入
  - 隐层
  - 隐层
  - …
  - 输出



$$\mathbf{H}_t^1 = f_1(\mathbf{H}_{t-1}^1, \mathbf{X}_t)$$
$$\mathbf{H}_t^j = f_j(\mathbf{H}_{t-1}^j, \mathbf{H}_t^{j-1})$$
$$\mathbf{O}_t = g(\mathbf{H}_t^L)$$

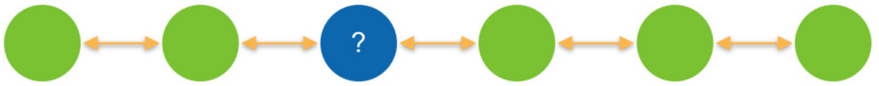
Let's try it

Bidirectional RNN : 一个前向RNN, 一个反向RNN, 合并两隐状态得输出.



$$\vec{\mathbf{H}}_t = \phi(\mathbf{X}_t \mathbf{W}_{xh}^{(f)} + \vec{\mathbf{H}}_{t-1} \mathbf{W}_{hh}^{(f)} + \mathbf{b}_h^{(f)}),$$
$$\overleftarrow{\mathbf{H}}_t = \phi(\mathbf{X}_t \mathbf{W}_{xh}^{(b)} + \overleftarrow{\mathbf{H}}_{t+1} \mathbf{W}_{hh}^{(b)} + \mathbf{b}_h^{(b)}),$$
$$\mathbf{H}_t = [\vec{\mathbf{H}}_t, \overleftarrow{\mathbf{H}}_t]$$
$$\mathbf{O}_t = \mathbf{H}_t \mathbf{W}_{hq} + \mathbf{b}_q$$

- 训练



- 推理

通常用于序列编码和给定双向上下文的观测估计，而不是预测未来